# One-Hot-Preserving QAOA for Flood Shelter Assignment: Pairwise Capacity Encoding and Angle Transfer

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***Abstract*—Flood shelter assignment is a small capacity-constrained assignment problem that is often proposed for quantum optimization. We study the Quantum Approximate Optimization Algorithm (QAOA) on instances from a flood digital twin by exact classical simulation. Capacity is encoded with approximate pairwise penalties instead of slack qubits; their minimiser was the true optimum in 93.3% of instances, so every sample is verified. At 12 qubits and depth 4, a ring-XY mixer that keeps each group assigned to one shelter placed a mean probability of 0.188 on the optimal plan over 5 optimizer seeds, against at most 0.048 for the transverse-field mixer at the same CNOT count per layer. One angle set per depth, chosen among angles trained on 30 development instances by mean optimal-plan probability, which needs their optima, was applied unchanged. It reached 0.306 on 30 held-out seeds against 0.163 for per-instance training and stayed ahead of it on 150 instances of 15 to 27 qubits (Holm $p \leq 0.027$). Rules needing no optima fell below per-instance training in leave-one-out tests, and transfer failed without a shared energy scale. Still, the optimal-plan probability fell about 2.9-fold per added group, and at 27 qubits only 0.053 of the probability lay on feasible plans. On 50 instances on the road networks of 6 Indian cities, fixed angles raised the chance that 32 verified samples contain the optimum. Exhaustive search solves the 12-qubit instances in about a millisecond; we claim no quantum advantage.**

***Index Terms*—QAOA, XY mixer, parameter transfer, QUBO, evacuation planning**

## I. Introduction

When a flood closes roads and fills shelters, an emergency officer must decide again which shelter each evacuation group should go to, trading off travel time, flood exposure and the seats left. As a binary program this is a capacity-constrained assignment problem, the kind that is routinely encoded as a quadratic unconstrained binary optimization (QUBO) problem [1], [2] and offered as an application for near-term quantum optimization [3], [4].

Prior proposals rarely ask three questions. Does the circuit respect the constraints or only penalise violations? The original QAOA [5] explores all $2^n$ bitstrings, most of which are not plans, while the quantum alternating operator ansatz [6] keeps the state in a feasible subspace [7]–[9]; but mixer studies treat equality constraints, and capacity is an inequality. Must the angles be retrained after every disruption, and how should a reusable set be chosen? And how do the results compare with classical methods and with a sampler followed by a feasibility check?

We address them on instances generated by the digital twin of a flood decision-support prototype. Our contributions are:

1) A shelter-assignment QUBO in which capacity is encoded by pairwise products instead of slack qubits, with a cost shift and scaling that place all instances on one energy range; we measure how often the encoding is exact and show that it admits infeasible zero-penalty plans.
2) A comparison of transverse-field, ring-XY, Grover and feasible-subspace mixers under mean-energy and conditional value-at-risk (CVaR) training over 5 optimizer seeds, with Holm-corrected tests and CNOT counts per layer.
3) An angle-transfer study with three selection rules, two of which need no optima, an ablation in which transfer failed without the shared energy scale, evidence that trained sets concentrate across instances, and a size test of 12-qubit angles up to 27 qubits.
4) The expected optimality gap of the best verified plan among $N$ samples, and a comparison with exhaustive search, a greedy rule and simulated annealing.

All results come from classical simulation; we make no claim of quantum speed-up.

## II. Related Work

*Formulations.* Lucas gave Ising forms for many NP-hard problems [1], and Glover *et al.* describe how assignment and capacity constraints become quadratic penalties, with slack variables for inequalities [2]. Flood shelter allocation has been solved by integer programming [10] and by quantum annealing [3].

*Mixers.* QAOA alternates a cost unitary and a mixing unitary for $p$ layers [5], [11]. Hadfield *et al.* generalised the mixer to preserve hard constraints [6]. Wang *et al.* analysed XY mixers for one-hot constraints [7], Fuchs *et al.* built constraint-preserving [8] and subspace mixers [12], and Grover mixers move the complexity into state preparation [9]. XY mixers have been paired with warm starts for routing [13] and adapted to rostering [14].

*Training and transfer.* Angles can be interpolated from depth $p$ to $p{+}1$ [15], CVaR objectives raise the probability of good samples [16], and warm starts use classical solutions [17]. The objective value at fixed angles concentrates across typical instances [18], optimal angles transfer between random

graphs [19], and for weighted MaxCut a rescaling of the weights lets one parameter set serve many instances [20]. On hardware, QAOA performance falls with size and depth [21], and practical advantage has not been shown [4].

*Positioning.* We do not propose a new mixer; the ring-XY mixer is that of [7]. The transfer studies [18]–[20] use unconstrained MaxCut with the standard mixer. We add a capacity inequality kept as a pairwise penalty next to a one-hot-preserving mixer, compare selection rules with and without optima, and evaluate on held-out instances and other road networks with a verifier in the loop.

## III. Problem Formulation

### A. Instances from the Digital Twin

The twin models a corridor with three population zones, nine junctions and three shelters joined by 22 roads. Every road carries a travel time, a flood-depth index and a congestion level drawn from a seed. Shelters have capacities of 120, 90 and 70 people with an initial occupancy of 15–45%, and groups have 8–22 people. Each instance applies the disruption used in the twin's routing benchmark: road R5 closes and Shelter A receives arrivals equal to half its capacity. For group $g$ and shelter $s$ we take the best open path under the twin's flood- and congestion-weighted edge cost and define

$$c_{gs} = T_{gs} + 5H_{gs}, \tag{1}$$

where $T_{gs}$ is travel time in minutes and $H_{gs}$ the mean flood index along the path. The pair $(g, s)$ is *forbidden* if no open path exists, the shelter has failed, or the group alone exceeds the remaining seats $R_s$. With $G$ groups, binary variables $x_{gs}$ give $n = 3G$ qubits. We write $m_g$ for the number of people in group $g$. A plan is feasible if every group has exactly one shelter, no forbidden pair is used and

$$\textstyle\sum_g m_g x_{gs} \le R_s \quad \text{for each shelter } s. \tag{2}$$

### B. QUBO with Pairwise Capacity

An inequality normally needs slack qubits [2]. We use a pairwise encoding instead. Let $\mathcal{O} = \{(g, h, s) : m_g + m_h > R_s\}$ be the pairs of groups that cannot share shelter $s$ and $\mathcal{F}$ the forbidden pairs. With shifted costs $c'_{gs} = c_{gs} - \min_{s'} c_{gs'}$ and penalty weight $\mu$,

$$E(\mathbf{x}) = \sum_{g,s} c'_{gs} x_{gs} + \mu\Big[\sum_g \Big(1 - \sum_s x_{gs}\Big)^2 + \sum_{(g,s)\in\mathcal{F}} x_{gs} + \sum_{(g,h,s)\in\mathcal{O}} x_{gs}x_{hs}\Big]. \tag{3}$$

*Remark 1:* With $\mu = 1 + \sum_g \max_s c'_{gs}$, every bitstring that breaks a one-hot, forbidden-pair or pairwise-capacity term has higher energy than every bitstring that breaks none of them, since $c'_{gs} \ge 0$ bounds the cost of a bitstring without violations by $\mu - 1$ and each violation adds at least $\mu$. The bound no longer holds for the reduced weights $\lambda\mu$ with $\lambda < 1$ used in the penalty sweep of Section VI-A.

On one-hot states the shift subtracts the constant $\sum_g \min_s c_{gs}$, so the optimal plan is unchanged. We divide the Hamiltonian by $\Delta = \mu - 1$, which maps the energies of all penalty-free plans into $[0, 1]$ for every instance. The pairwise encoding misses overloads by three or more groups whose pairs all fit, so a penalty-free plan can still be infeasible, and every sampled plan is re-checked exactly before use.

## IV. QAOA Variants and Training

For depth $p$ and angles $\boldsymbol{\theta} = (\boldsymbol{\gamma}, \boldsymbol{\beta})$, QAOA prepares

$$|\psi(\boldsymbol{\theta})\rangle = \prod_{k=1}^{p} e^{-i\beta_k B}\, e^{-i\gamma_k \hat{E}/\Delta}\, |\psi_0\rangle, \tag{4}$$

where $\hat{E}$ is diagonal with entries $E(\mathbf{x})$. Substituting $x_q = (1 - Z_q)/2$ turns (3) into an Ising Hamiltonian with one $R_Z$ rotation per qubit and one $ZZ$ rotation, two CNOTs, per non-zero coupling.

*Transverse-field (X) mixer.* $B = \sum_q X_q$ and $|\psi_0\rangle = |+\rangle^{\otimes n}$; constraints act only through the penalties. A cost layer has 12 one-hot couplings and on average 2.0 capacity couplings, 28 CNOTs; the mixer needs none.

*Ring-XY mixer.* For each group $g$ with qubits $a_g, b_g, c_g$,

$$B_{\mathrm{XY}} = \sum_g \sum_{(u,v)\in\{(a_g,b_g),(b_g,c_g),(c_g,a_g)\}} \tfrac{1}{2}\big(X_u X_v + Y_u Y_v\big). \tag{5}$$

Each term swaps amplitude between $|01\rangle$ and $|10\rangle$, so a one-hot group stays one-hot [7]. The start state $|W_3\rangle^{\otimes G}$, with $|W_3\rangle = (|100\rangle + |010\rangle + |001\rangle)/\sqrt{3}$, is the uniform superposition of the $3^G$ one-hot plans [22]. The one-hot penalty is then constant and dropped; the three ring terms are applied in a fixed order. Each XY rotation compiles to two CNOTs [23], so a layer has 28 CNOTs, the same count as the X variant. We prepare $|W_3\rangle$ with an $R_Y$ rotation, a controlled Hadamard (one CNOT) and two CNOTs, which we checked numerically; this adds 12 CNOTs once.

*Grover mixer.* $B_{\mathrm{GM}} = |\psi_0\rangle\langle\psi_0|$ with the same start state, so $e^{-i\beta B_{\mathrm{GM}}} = I + (e^{-i\beta} - 1)|\psi_0\rangle\langle\psi_0|$ [9]. A layer needs the inverse preparation, a phase on $|0\rangle^{\otimes n}$ controlled by all $n$ qubits, and the preparation. With the textbook ancilla construction [24] the controlled phase takes $2(n-2)$ six-CNOT Toffolis, one controlled phase and $n - 2 = 10$ ancillas (122 CNOTs), so a Grover layer has 150 CNOTs.

*Feasible-subspace mixer.* As a bound, not a practical circuit, a Grover mixer starts from the uniform superposition of the feasible plans, which needs the feasible set [9], [12].

*Objectives and training.* We minimise either $\langle\hat{E}\rangle/\Delta$ or the CVaR with $\alpha = 0.1$ [16]. Sorting bitstrings by energy, $E_{(1)} \le E_{(2)} \le \dots$, with probabilities $\pi_{(k)}$ and cumulative mass $\Pi_k = \sum_{j\le k} \pi_{(j)}$,

$$\mathrm{CVaR}_\alpha = \frac{1}{\alpha} \sum_k E_{(k)}\, \min\big(\pi_{(k)},\ \max(0,\ \alpha - \Pi_{k-1})\big). \tag{6}$$

Every variant is trained from scratch with the same procedure: COBYLA [25] with at most 250 evaluations per start, depth by depth. At each depth $p$ we start from two random points ($\gamma_k \sim U(0, \pi)$, $\beta_k \sim U(0, \pi/2)$) and, for $p > 1$, from the optimum of depth $p - 1$ interpolated to $p$ layers [15], and

keep the start with the lowest objective. Training never uses the optimal plan.

### A. Angle Transfer

Since the scaled Hamiltonians of all instances share the energy range $[0, 1]$, we test whether trained angles carry over. Let $\Theta_p = \{\boldsymbol{\theta}_j^{(p)} : j \in \mathcal{D}\}$ be the angles trained on the development instances $\mathcal{D}$. We fix one set $\boldsymbol{\theta}_p^\star \in \Theta_p$ by one of three rules:

$$\boldsymbol{\theta}_p^\star = \arg\max_{\boldsymbol{\theta} \in \Theta_p} \ \frac{1}{|\mathcal{D}|} \sum_{j \in \mathcal{D}} P_{\mathrm{opt}}(j; \boldsymbol{\theta}), \tag{7}$$

or the set with the lowest mean of $\langle \hat{E} \rangle / \Delta$, or with the lowest mean CVaR, over $\mathcal{D}$. Rule (7) needs the exact optimum of every development instance, from exhaustive search; it is an oracle on the development set, though it uses nothing from the instances it is applied to. The other two rules use only quantities a device can estimate. We choose among whole trained sets because their coordinate-wise median, which mixes different basins of the periodic angle space, gave $P_{\mathrm{opt}} = 0.001$ in a leave-one-out test.

## V. Experimental Setup

*Instances.* The development set has 30 instances from twin seeds $42 + 17i$, $i = 0, \ldots, 29$, with the first four groups ($G = 4$, $n = 12$). All are feasible, with on average 45.0 feasible plans among the 81 one-hot plans, and in 12 of them capacity changes the optimal plan (*capacity-binding*). The held-out set has 30 further seeds, $i = 30, \ldots, 59$, of the same disruption pattern, used only for the final transfer test. We also built 60 instances (50 feasible, 22 capacity-binding) on the OpenStreetMap road graphs of 6 flood-prone Indian cities, with each city's mapped zones, shelters and road flood-exposure proxy. Demand is not mapped, so group sizes and seats follow the twin; the disruption closes the most flood-exposed road on any zone–shelter path and halves the first shelter's seats.

*Metrics.* Simulations store all $2^n$ amplitudes in double precision, with fixed seeds. $P_{\mathrm{opt}}$ is the probability of measuring an optimal plan and $P_{\mathrm{feas}}$ that of a feasible plan. A sampler draws $N$ plans and returns the cheapest one that passes an exact feasibility check. It returns the optimum whenever the optimum is among the samples, which has probability success@$N = 1 - (1 - P_{\mathrm{opt}})^N$, assuming independent shots. We also report the expected relative gap $(c - c^\star)/c^\star$ of the returned plan, given that a feasible plan was drawn, for $N \in \{8, 32, 128\}$, computed exactly from the output distribution. An evaluation is one estimate of the training objective (on hardware, a batch of circuit runs).

*Statistics.* Paired comparisons use the two-sided Wilcoxon signed-rank test on per-instance values, also for success@$N$. We apply the Holm correction within each family of tests: the four depths of one mixer comparison, the 7 rows of Table II, the four city comparisons and the 10 size comparisons. Effect sizes are median paired differences. Intervals for variants run with 5 optimizer seeds use a two-level bootstrap that resamples instances and then one seed per instance. Because the city instances cluster by city, we also apply a sign test to the six per-city mean differences.

*Baselines.* Uniform sampling over the 81 one-hot plans ($P_{\mathrm{opt}} = 0.0169$) and over the feasible plans ($P_{\mathrm{opt}} = 0.049$); exhaustive search; a greedy rule that places the largest group first in its cheapest shelter with seats left; and simulated annealing (SA) [26] with 20 restarts of 200 sweeps, either with bit flips on (3) or with moves that reassign one group, which keep the plan one-hot. For the X mixer we swept the penalty $\lambda\mu$ with $\lambda \in \{0.25, 0.5, 1, 2\}$.

TABLE I
Mixers and objectives at depth 4 (30 development instances, 12 qubits)

| Variant | $P_{\mathrm{opt}}$ | 95% CI | $P_{\mathrm{feas}}$ | s@32 % | Evals | CNOT |
|---|---|---|---|---|---|---|
| X, mean | 0.040 | 0.028–0.055 | 0.569 | 60.8 | 2,091 | 28 |
| XY, mean | 0.188 | 0.127–0.257 | 0.755 | 82.5 | 2,510 | 28 |
| XY, CVaR | 0.139 | 0.117–0.166 | 0.664 | 97.7 | 1,356 | 28 |
| Grover, mean | 0.139 | 0.121–0.157 | 0.815 | 96.9 | 2,033 | 150 |
| Grover, CVaR | 0.117 | 0.106–0.127 | 0.563 | 96.2 | 918 | 150 |
| Feasible subspace | 0.444 | 0.385–0.511 | 1.000 | 100.0 | 2,039 | n/a |

X, XY: mean over 5 seeds, CI by two-level bootstrap. Evals: objective evaluations per instance, depths 1–4. CNOT: per layer, all-to-all connectivity.

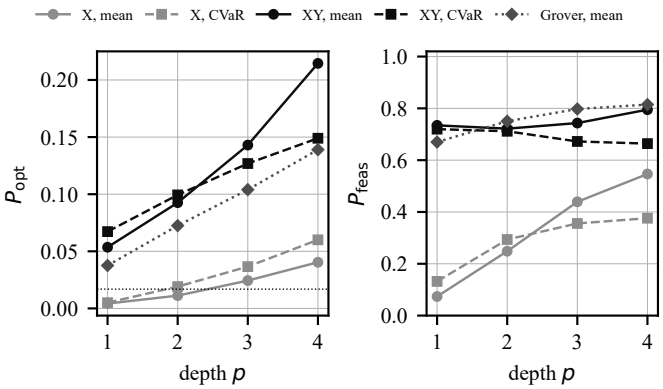



Fig. 1. Probability of the optimal plan (left, different $y$-range) and of any feasible plan (right) against depth, optimizer seed 0. Dotted: uniform one-hot sampling.

## VI. Results

### A. Mixers, Objectives and Penalties

Table I and Fig. 1 compare the mixers. Over 5 seeds the XY mixer reached $P_{\mathrm{opt}} = 0.188 \pm 0.020$ at $p = 4$ (seed means 0.168–0.215), against 0.035–0.048 for the X mixer. It was better on 25–29 of 30 instances at every depth (Holm-adjusted $p \leq 8.0 \times 10^{-6}$), and the ranking held for every seed. Its feasibility stayed between 0.722 and 0.795 at every depth, whereas the X mixer began with $P_{\mathrm{feas}} = 0.074$ and spent its layers recovering feasibility.

The Grover mixer reached 0.139 at $p = 4$, and its difference from XY was not significant at any depth (Holm-adjusted $p \geq 0.19$). The data lean towards XY: at $p = 4$ it was ahead on 20 of 30 instances, with a mean difference of 0.076

(paired bootstrap interval 0.018–0.141) and an unadjusted $p = 0.047$. The Grover mixer spread probability more evenly over instances, so its success@32 was higher, but a layer costs 150 CNOTs and 10 ancillas, against 28 CNOTs for ring-XY.

CVaR training lowered the XY mixer's mean $P_{\text{opt}}$ but raised success@32 from 88.4% to 98.1% (seed 0): the worst instance, like all the worst ones capacity-binding, rose from 0.002 to 0.074.

A smaller penalty helped the X mixer only partly: lowering the weight to $\lambda = 0.25$ raised its $P_{\text{opt}}$ by 2.6× to 0.104 at a feasibility of only 0.401, still about half of the XY result. As a bound on the cost of a capacity penalty, a mixer restricted to feasible plans reached $P_{\text{opt}} = 0.444$ at $p = 4$, 9.0 times uniform sampling over the same feasible plans, and beat XY on 26 of 30 instances ($p = 4.4 \times 10^{-6}$).

### B. Angle Transfer to Held-Out Seeds

Table II gives the held-out test. For the XY mixer with the mean objective, the set chosen by (7) reached $P_{\text{opt}} = 0.306$ against 0.163 for per-instance training with 2,496 evaluations (better on 24 of 30, median difference 0.195, Holm $p = 8.5 \times 10^{-4}$). The oracle-free sets also beat per-instance training on these seeds: 0.264 for the energy rule ($p = 3.8 \times 10^{-3}$) and 0.234 for the CVaR rule ($p = 0.039$). Leave-one-out on the development set tells a different story: the energy rule gave 0.082 and the CVaR rule 0.164, both below the 0.215 of per-instance training, while rule (7) gave 0.292. The same held for CVaR angles chosen by CVaR (0.250 on held-out seeds, 0.095 in leave-one-out against 0.149). An oracle-free choice therefore gave no reliable gain, while rule (7) gained in both tests. Transfer did not rescue the X mixer (0.036).

Part of the gap is a mismatch of objectives. Per-instance training reached a lower mean $\langle \hat{E} \rangle / \Delta$ than the fixed set (0.300 against 0.530, lower on 24 of 30), but low energy did not imply a high $P_{\text{opt}}$: across the 30 trained sets the rank correlation between mean energy and mean $P_{\text{opt}}$ on the development set was only 0.33. Stronger optimizers stayed below: the best of five COBYLA seeds by training objective reached 0.203 with 12,485 evaluations per instance, and L-BFGS-B 0.225. COBYLA started from the fixed angles did improve on them (0.385 with 814 evaluations, $p = 1.9 \times 10^{-3}$).

Why does a fixed set work? Performance concentrates across instances: a trained set's mean $P_{\text{opt}}$ on the development set predicted its mean on the held-out seeds with rank correlation 0.98. Only 7 of the 30 trained sets beat per-instance training there (median 0.062), so the choice matters. The gain came mostly from capacity-binding instances: on the 16 binding seeds per-instance training reached 0.106 and the fixed set 0.312, on the others 0.228 and 0.298. We repeated training, selection and the held-out test without the scaling and with division by the standard deviation of the penalty-free energies. Without scaling the fixed angles reached only 0.029, although per-instance training still reached 0.270; with the other normalisation they reached 0.282, close to our 0.306. Transfer needs a common energy scale, not our particular one. The one-time cost, 75,392 training and 3,600 selection evaluations, equals per-instance training on about 32 disruptions.

### C. Sampling with a Feasibility Check

Table III shows what the sampler of Section V returns. With 8 samples, the fixed XY angles came within 0.09% of the optimal cost on average, per-instance training within 1.49% and uniform one-hot sampling within 6.7%. With 32 samples every XY variant drew a feasible plan with probability 100.0% to one decimal, and with 128 samples per-instance training was within 0.10%. Since many plans cost close to the optimum, success@$N$ is the stricter measure.

### D. Transfer to Real Road Networks

Table IV gives the city test, with angles fixed on the twin. Pooled over the 50 instances, the set chosen by (7) reached $P_{\text{opt}} = 0.202$ against 0.213 for per-instance training (Holm $p = 0.74$) and the energy-rule set 0.216 ($p = 0.38$). It led in only 3 of 6 cities (sign test $p = 1.00$), so there was no gain in $P_{\text{opt}}$. Per-instance training left 8 instances with $P_{\text{opt}} < 0.01$, the fixed set 1, and success@32 rose from 74.9% to 89.9% (Holm $p = 0.042$) and to 96.7% for the energy rule ($p = 9.7 \times 10^{-5}$), in all 6 cities (sign test $p = 0.03$).

### E. Larger Instances and Gate Noise

The ring-XY circuit never leaves the $3^G$ one-hot amplitudes, so we simulated it exactly in that subspace; on 40 random 12- and 15-qubit circuits it agreed with the full statevector to $1.6 \times 10^{-16}$. We took 30 new twin instances for each of $G = 5, \ldots, 9$ groups (15 to 27 qubits) with the same disruption, applied the 12-qubit sets of rule (7) and of the energy rule unchanged, and trained every instance with the procedure above (Table V). The set of rule (7) beat per-instance training at every size (Holm $p \leq 0.027$ over 10 tests). The energy-rule set won on at least 22 of 30 instances at every size but was significant only at 21 and 24 qubits. Per-instance training (about 2,438 evaluations) fell below $P_{\text{opt}} = 0.01$ on at least 28 of 30 instances from 21 qubits on.

The absolute values temper this result. The fixed set's mean $P_{\text{opt}}$ fell by a factor of about 2.9 per added group, to 0.003 at 27 qubits, still a median 16 times uniform one-hot sampling; from 21 qubits on it fell below uniform sampling on up to 4 instances. Capacity is the main cause. From 24 qubits on every instance was capacity-binding, and the fixed circuit put at most 0.053 of its probability on feasible plans, since the one-hot mixer sees capacity only through the penalty. Most samples then fail the verifier.

We simulated gate noise by quantum trajectories (200 runs per circuit) on 10 development instances: each two-qubit gate, including four per $W_3$ preparation, fails with probability $\epsilon$ and applies one of the 15 non-identity two-qubit Paulis. As each $ZZ$ and XY gate compiles to two CNOTs, $\epsilon = 0.01$ means a CNOT error near 0.5%, close to the 0.36% (isolated) and 0.62% (simultaneous) two-qubit errors of a superconducting processor [27]. At $\epsilon = 0.01$ and $p = 4$ the XY mixer kept $P_{\text{opt}} = 0.105$, 3.2× the X mixer's 0.033. A global depolarising

TABLE II
ANGLES AT $p = 4$ ON THE 30 HELD-OUT SEEDS: PER-INSTANCE TRAINING AGAINST ONE FIXED ANGLE SET CHOSEN ON THE DEVELOPMENT SET

| Variant | Angles | Needs optima | Held-out $P_{\text{opt}}$ | Median paired diff. | Wins of 30 | Holm $p$ | Leave-one-out $P_{\text{opt}}$ (dev.) | s@32 % |
|---|---|---|---|---|---|---|---|---|
| XY, mean | per instance | no | 0.163 | | | | 0.215 | 79.7 |
| | best of 5, min. energy | no | 0.203 | | | | 0.245 | 84.9 |
| | L-BFGS-B | no | 0.225 | | | | | 76.9 |
| | fixed, max. $P_{\text{opt}}$ | yes | 0.306 | +0.195 | 24 | $8.5 \times 10^{-4}$ | 0.292 | 100.0 |
| | fixed, min. energy | no | 0.264 | +0.131 | 24 | $3.8 \times 10^{-3}$ | 0.082 | 99.9 |
| | fixed, min. CVaR | no | 0.234 | +0.102 | 23 | 0.039 | 0.164 | 99.8 |
| XY, CVaR | per instance | no | 0.138 | | | | 0.149 | 97.9 |
| | fixed, max. $P_{\text{opt}}$ | yes | 0.250 | +0.070 | 29 | $6.5 \times 10^{-8}$ | 0.259 | 99.8 |
| | fixed, min. CVaR | no | 0.250 | +0.070 | 29 | $6.5 \times 10^{-8}$ | 0.095 | 99.8 |
| X, mean | per instance | no | 0.028 | | | | 0.040 | 50.3 |
| | fixed, max. $P_{\text{opt}}$ | yes | 0.036 | +0.010 | 20 | 0.039 | 0.039 | 63.6 |
| | fixed, min. energy | no | 0.034 | +0.006 | 21 | 0.064 | 0.034 | 65.8 |

Fixed sets vs. per-instance training of the same variant (Wilcoxon signed-rank, Holm over 7 rows). Per-instance rows: development-set mean in the leave-one-out column.

TABLE III
BEST VERIFIED PLAN AMONG $N$ SAMPLES (HELD-OUT SEEDS, $p = 4$): MEAN GAP TO THE OPTIMUM AND CHANCE OF DRAWING IT (%)

| | $N = 8$ | | $N = 32$ | | $N = 128$ | |
|---|---|---|---|---|---|---|
| Sampler | gap | opt. | gap | opt. | gap | opt. |
| uniform one-hot | 6.71 | 14.1 | 2.17 | 44.7 | 0.22 | 88.3 |
| uniform feasible | 4.17 | 26.8 | 0.96 | 67.7 | 0.05 | 96.4 |
| XY, per instance | 1.49 | 55.0 | 0.48 | 79.7 | 0.10 | 92.8 |
| XY, fixed (max. $P_{\text{opt}}$) | 0.09 | 93.9 | 0.00 | 100.0 | 0.00 | 100.0 |
| XY, fixed (min. energy) | 0.22 | 87.7 | 0.00 | 99.9 | 0.00 | 100.0 |

TABLE IV
OPENSTREETMAP CITY INSTANCES (XY MIXER, $p = 4$)

| | | $P_{\text{opt}}$ | | | s@32 % | | |
|---|---|---|---|---|---|---|---|
| City | inst. | trained | fix-O | fix-E | trained | fix-O | fix-E |
| Delhi | 10 | 0.142 | 0.184 | 0.164 | 76 | 91 | 94 |
| Chennai | 10 | 0.137 | 0.165 | 0.215 | 66 | 92 | 96 |
| Mumbai | 10 | 0.212 | 0.258 | 0.248 | 95 | 100 | 100 |
| Kolkata | 5 | 0.341 | 0.220 | 0.247 | 64 | 84 | 97 |
| Guwahati | 10 | 0.229 | 0.155 | 0.207 | 75 | 75 | 96 |
| Patna | 5 | 0.354 | 0.274 | 0.248 | 61 | 99 | 99 |
| All | 50 | 0.213 | 0.202 | 0.216 | 75 | 90 | 97 |

fix-O: set chosen by max. mean $P_{\text{opt}}$ (7); fix-E: by min. mean energy.

model, which returns the ideal state with probability $(1-\epsilon)^{N_2}$ for $N_2$ two-qubit gates, gave a lower 0.086, since many Pauli errors barely change the output.

### F. Encoding Exactness and Classical Baselines

In all 30 development instances some infeasible one-hot plan had zero penalty: three groups whose pairs fit overloaded a shelter together. Such plans usually cost more than the optimum, so the minimiser of (3) was the true optimum in 93.3% of instances. Exhaustive search took 0.90 ms per instance. The greedy rule found a feasible plan in 100.0% of instances and the optimum in 93.3%. Bit-flip SA ended at the optimum in 80.0% of instances and one-hot SA in 93.3%, the encoding's limit; both visited the optimum every time, so keeping the cheapest verified plan found it in 100%.

TABLE V
FIXED 12-QUBIT ANGLES ON 30 TWIN INSTANCES PER SIZE $n$ (QUBITS), XY MIXER, $p = 4$; $P_{\text{feas}}$ OF FIX-O

| | $P_{\text{opt}}$ | | | | | | s@32 % | |
|---|---|---|---|---|---|---|---|---|
| $n$ | trained | fix-O | Holm $p$ | fix-E | Holm $p$ | $P_{\text{feas}}$ | trained | fix-O |
| 15 | 0.1110 | 0.1924 | $4.0 \times 10^{-3}$ | 0.1470 | 0.083 | 0.50 | 56.7 | 99.6 |
| 18 | 0.0415 | 0.0670 | 0.027 | 0.0505 | 0.083 | 0.36 | 31.2 | 84.4 |
| 21 | 0.0037 | 0.0211 | $2.0 \times 10^{-5}$ | 0.0139 | $1.5 \times 10^{-3}$ | 0.13 | 9.3 | 42.5 |
| 24 | 0.0016 | 0.0047 | $4.0 \times 10^{-3}$ | 0.0034 | $5.7 \times 10^{-3}$ | 0.04 | 3.9 | 12.4 |
| 27 | 0.0013 | 0.0026 | 0.011 | 0.0012 | 0.083 | 0.05 | 3.5 | 7.6 |

## VII. DISCUSSION AND LIMITATIONS

Three findings matter for hardware experiments. First, the one-hot constraint belongs in the mixer: at the same CNOT count, ring-XY beat the transverse-field mixer on almost every instance and seed, and a smaller penalty did not close the gap. Capacity is the open part: the feasible-subspace mixer roughly doubled $P_{\text{opt}}$ at 12 qubits but needs the feasible set, and once capacity bound on every instance ring-XY put almost all its probability on infeasible plans. The next step is a capacity-aware mixer or a slack encoding. Second, transfer needs a common energy scale and a good selection rule: with development-set optima one fixed set per depth beat per-instance training, for a one-time cost of 78,992 evaluations, but without optima the gain was not reliable. Third, infeasible penalty-free plans exist, so every sample must be verified.

The study has limits. All twin instances share one disruption pattern; other closures, failed shelters and capacity cuts were not tested, and the city instances use real roads but sampled demand. The noise model charges errors per logical gate, ignores single-qubit and correlated errors, omits the Grover mixer and was run at 12 qubits only; no circuit was compiled for hardware. $P_{\text{opt}}$ values depend on the training budget, and the Grover test cannot show equivalence.

## VIII. CONCLUSION

On flood shelter assignment, the ring-XY mixer gave QAOA a significantly higher $P_{\text{opt}}$ than the transverse-field mixer at equal CNOT count. Angles fixed with development-set optima

beat per-instance training from 12 to 27 qubits; rules without optima were not reliable. Capacity, kept as an approximate penalty, needs a verifier and dominated at scale. The results guide hardware experiments and show no present-day advantage.

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